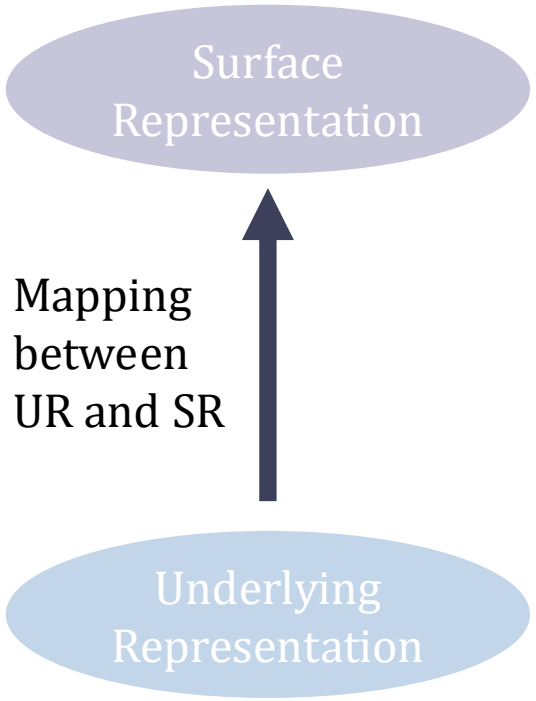
Learning Interacting Phonological Processes:
An Artificial Language Learning Approach

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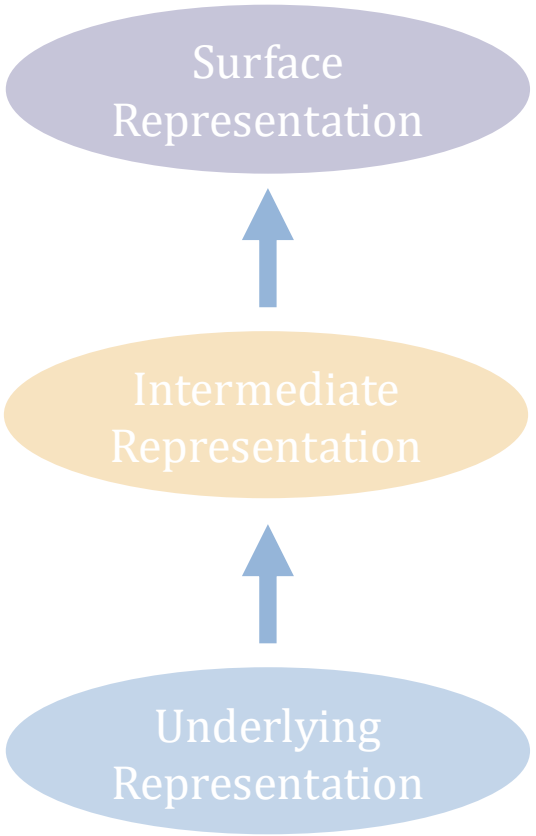
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Overview



- It is widely accepted in generative phonology that there is a mapping between the abstract underlying representation (UR) and surface representation (SR).
- Learners must infer the processes underlying observed phonological patterns.

Overview



- This work focuses on learning pair of interacting processes.
  - Specifically, the learning problem is: when learners are given surface forms (paired with their underlying forms) resulting from application of two processes, how can they learn the individual processes and the ordering?

Interaction

- Phonological processes can interact when they share similar targets or triggers, such that the application of one process may influence another.

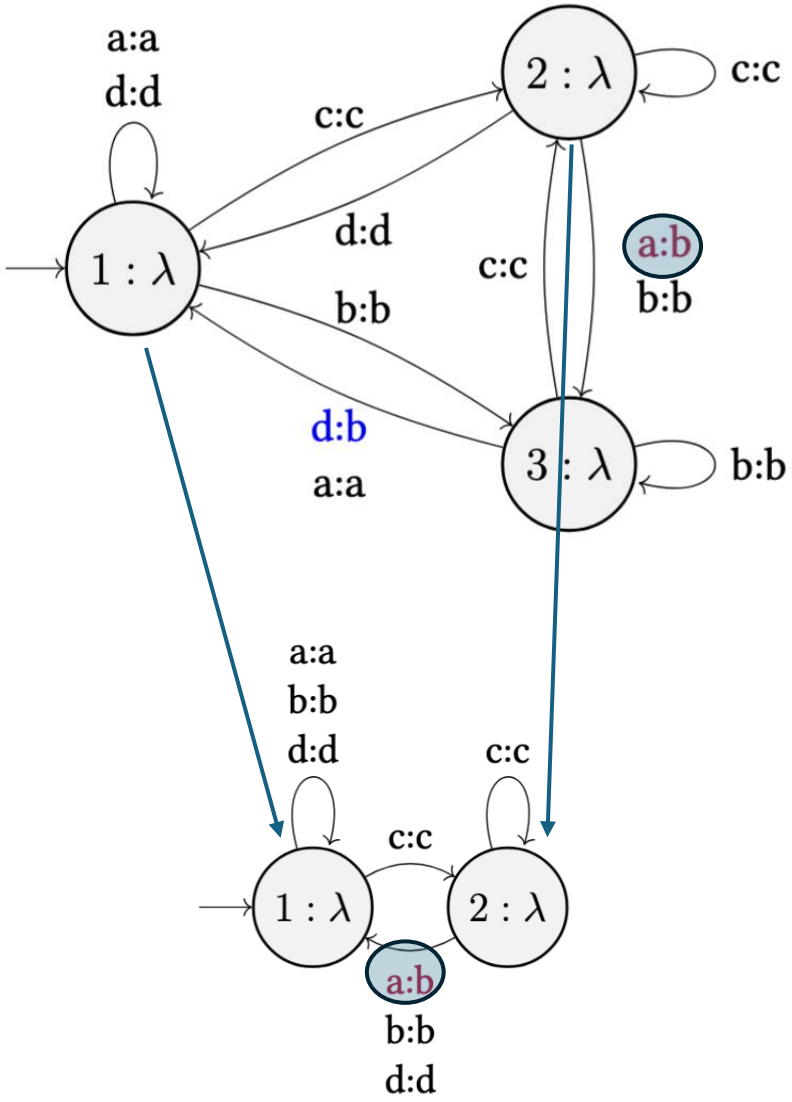
- Many Chinese dialects exhibit rich tone sandhi systems operating over adjacent tones.
  - Rule defined over two-syllable domains often extend to longer sequences, resulting in interactions
  - Example (Changting dialect) — five lexical tones: H, M, L, F, R

    - **RM Rule:** R → H / _M e.g., *han*^{R}*leng*^{M} → *han*^{H}*leng*^{M} 'cold'
    - **LF Rule:** L → M / _F e.g., *dai*^{L}*biao*^{F} → *dai*^{M}*biao*^{F} 'represent' (Chen, 2004)

    wen^{R}xue^{L}shi^{F} → wen^{H}xue^{M}shi^{F}
    /RLF/ → R**M**F → [**H**MF] (LF feeds RM)

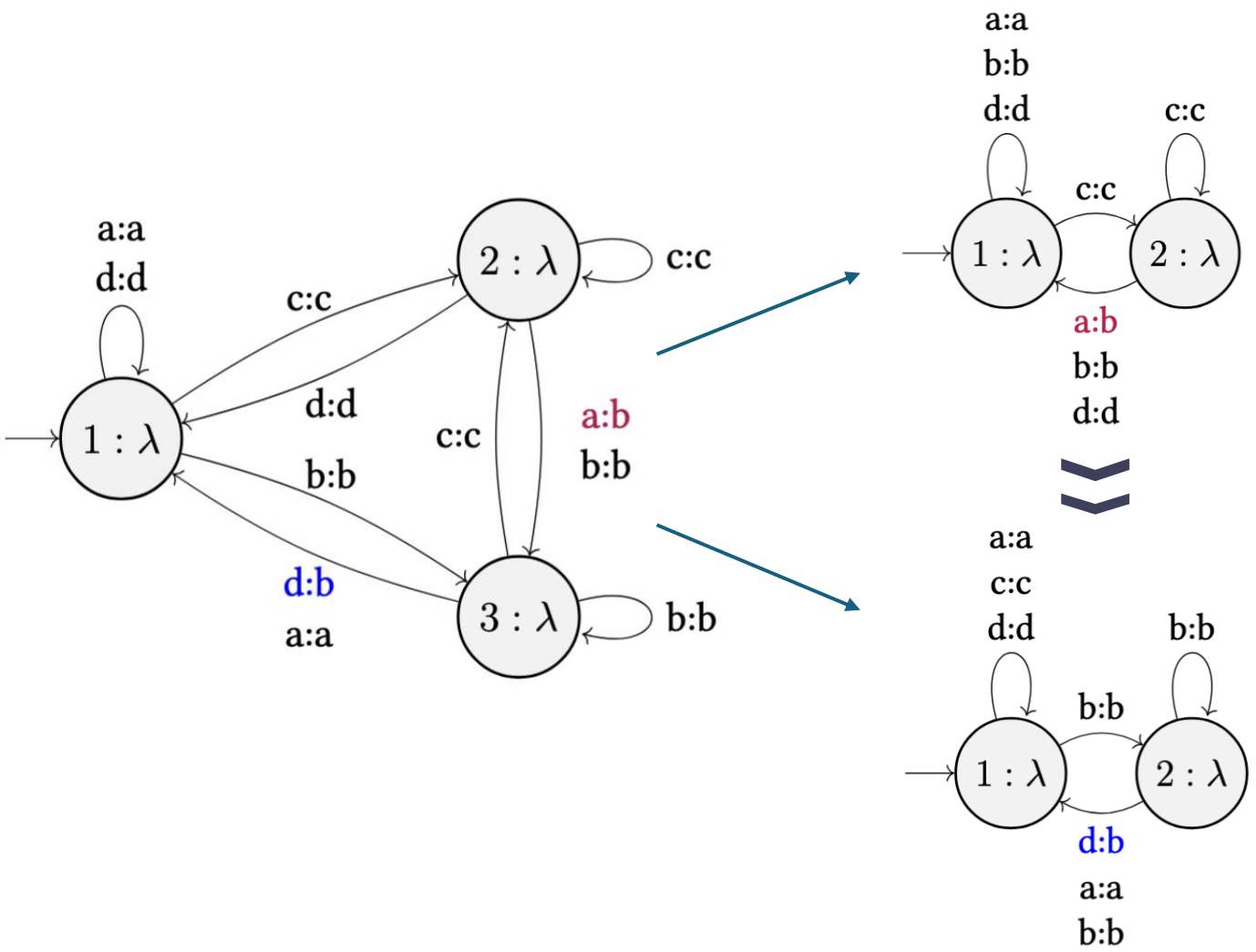
- The tone sandhi patterns give rise to both transparent (feeding &bleeding) and opaque (counterfeeding and counterbleeding) interaction

Overview – Proposed Learning Procedure



- Learning assumption: The learner is given underlying–surface pairs.
  - Including pairs demonstrating the individual processes and interactions.
- First, the mapping is represented as a whole by an interaction transducer generated by OSTIA.
- Next, the learner attempts to *decompose* the combined mapping into two individual phonological processes and infer their relative ordering

Representation of Interaction



- A single transducer can represent the entire combined mapping.
- Why decompose it into individual processes and their ordering?
- Are learners sensitive to the internal structure of an interaction?

Learning Biases

- Transparency Bias (Kiparsky, 1971)
  - Preference for transparent interactions, where surface forms are consistent with the application of the individual processes.
  - Favors feeding and bleeding over counterfeeding and counterbleeding.

Prickett (2019)

- Investigates how learning bias affect learning using Artificial Language Learning paradigm.
- Four artificial languages differ in interaction type.

Palatalization: /t, d/ → [tʃ, dʒ] / __ i
e.g., /inat-i/ → [inatʃ-i]

Vowel deletion: $V_1V_2 \rightarrow V_2$
e.g., /inak-i-a/ → [inak-a]

Feeding: /inat-a-i/ → [inatʃ-i]
Bleeding: /inat-i-a/ → [inat-a]
Counterfeeding: /inat-a-i/ → [inat-i]
Counterbleeding: /inat-i-a/ → [inatʃ-a]

Prickett (2019)

- Each language contained four different trial types:

Faithful	Deletion-only	Palatalization-only	Interacting
/t-a/ → [t-a]	/k-a-i/ → [k-i]	/t-i/ → [tʃ-i]	/t-a-i/ → [tʃ-i]

Prickett (2019)

- Training: Two-alternative forced-choice task (correct vs. foil pronunciation), with feedback.

Faithful	Deletion-only	Palatalization-only	Interacting
/ta/ [ta] vs. *[tʃa]	/kai/ [ki] vs. * [kai]	/ti/ [tʃi] vs. *[ti]	/tai/ [tʃi] vs. *[ti]

- Test: Novel stems, no feedback; accuracy analyzed separately by trial type.

Prickett (2019)

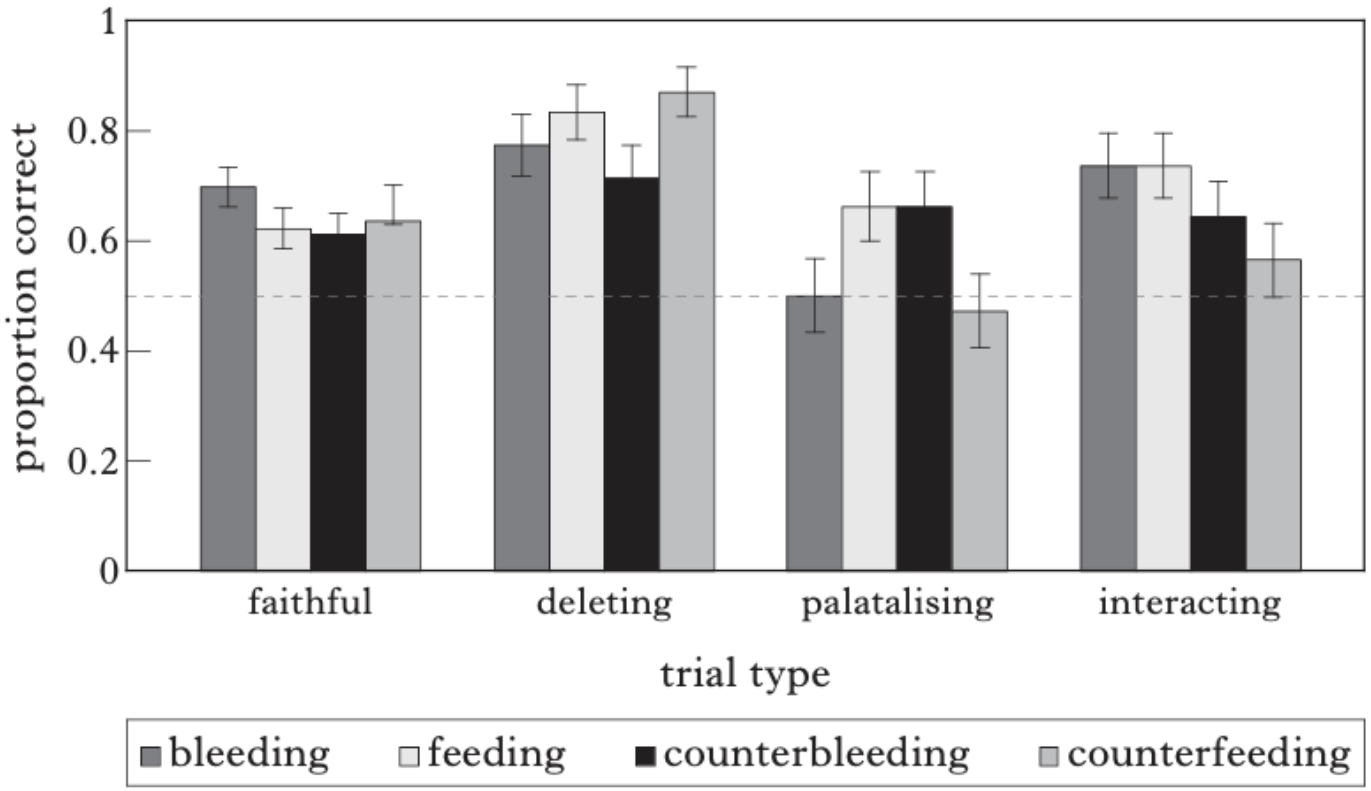


Figure 1

Average proportion of correct answers in the testing phase, by pattern type and trial type. See Table I for explanations of each trial type. Error bars show 95% confidence intervals calculated over all datapoints in each condition.

Research Question & Hypothesis

- **Observation**: A *transparency advantage* emerges in interacting trials.
- **Hypothesis**: The Transparency Bias may reflect learners' *sensitivity to the component processes* and their *ordering*.

- **Experimental Questions**:
  - Experiment 1: Does the Transparency Bias generalize to interactions involving length-preserving processes?
  - Experiment 2: Does the availability of component-process evidence influence interaction learning and the Transparency Bias?

Pilot

- Establish whether participants can learn and generalize individual phonological processes using the current artificial language learning paradigm.
- Participants are exposed to two independent processes: palatalization and either vowel raising or vowel lowering
  - Palatalization: /t, d/ → [tʃ, dʒ] / __ i
  - VR: /e/ → [i] / __ m
  - VL: /i/ → [e] / __ m
- Pre-training: participants are instructed on the meaning of the suffixes.
- In Training, participants choose between two possible pronunciations of a derived form, with feedback.
- The test phase examines generalization to novel stems and suffixes.



Play

Press SPACEBAR to move on

Play Original



Play 1

Play 2

Choose 1

Choose 2

Play Original

Incorrect :(

correct 1



Play

Press SPACEBAR to move on

Play Original

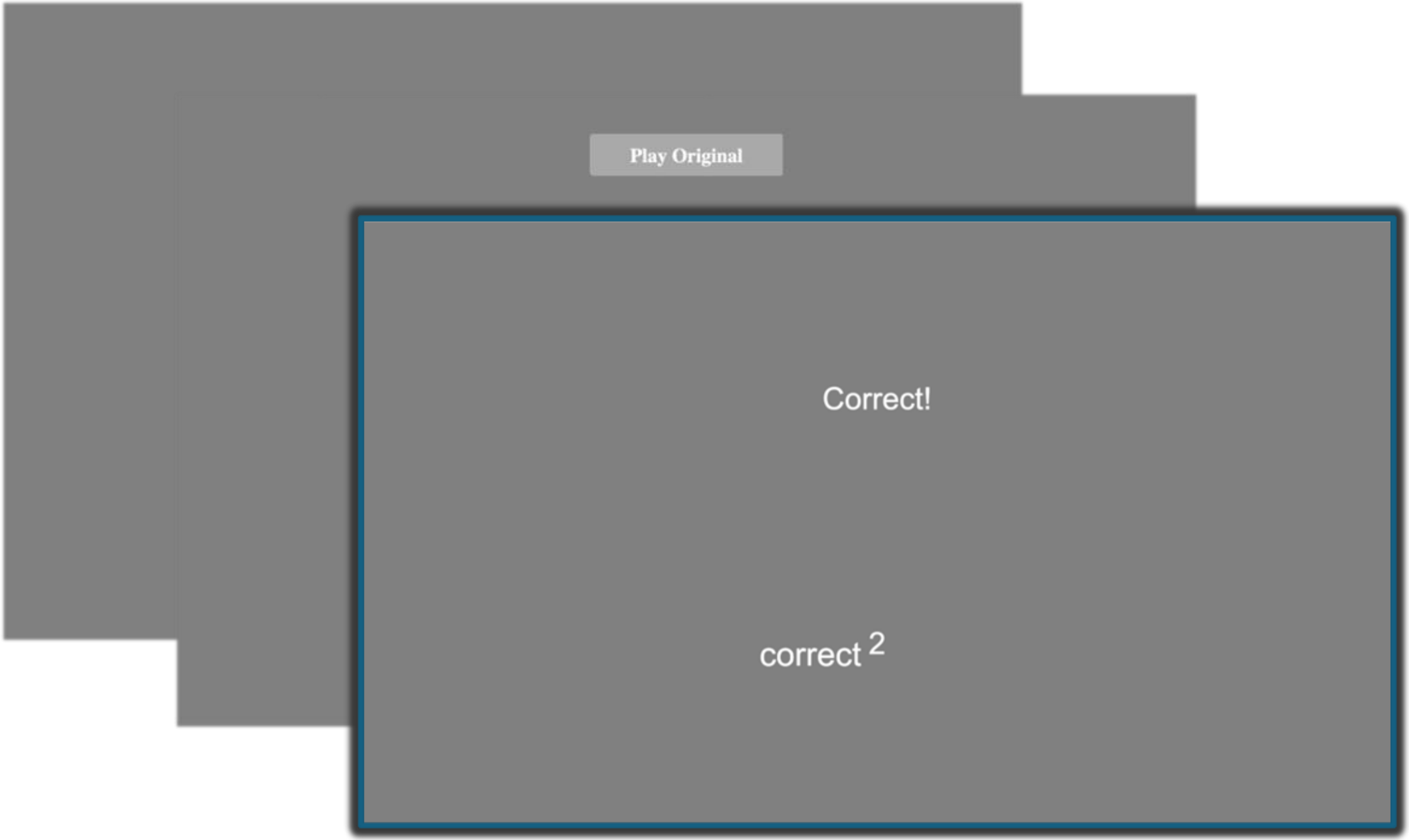


Play 1

Play 2

Choose 1

Choose 2


Play Original
Correct!
correct 2

Planned Experiment 1

- Learning Biases with Component and Interaction Evidence
- Goal: Investigate the learnability of different interaction types when learners receive evidence for both the individual processes and their interaction.
- Proposed Design:
  - Four conditions: feeding, bleeding, counterfeeding, and counterbleeding.
  - Training includes evidence for both individual processes and their interaction.
  - Processes: palatalization + vowel raising or lowering.
- Prediction:
  - Higher accuracy on interacting trials for feeding and bleeding than for counterfeeding and counterbleeding.

Planned Experiment 2

- Representation of interaction
- Goal: Investigate whether separate evidence for component processes affects interaction learning.
- Manipulations:

Component + Interaction	**Interaction Only**
Training provides evidence for individual processes and their interaction.	Training provides interacting mappings without separate component-process examples.

- Test:
  - Generalization to new interacting forms
  - Generalization to contexts involving individual processes.
- Hypothesis:
  - Component-process evidence facilitates generalization to novel interacting forms.
  - The availability of component-process evidence modulates the Transparency Bias.

Conclusion

- Computational perspective
  - The learner decomposes a combined mapping into individual processes and their ordering.
- Human learning
  - The Transparency Bias raises the possibility that learners are sensitive to the internal structure of phonological interactions.
- Experimental approach
  - Experiment 1: Test whether the Transparency Bias *generalizes* to length-preserving processes.
  - Experiment 2: Investigate how component-process evidence affects interaction learning and the Transparency Bias.

Thank you!